

Apply filters to SQL queries

Project description

Databases contain large amounts of data organized by tables. To access database data efficiently, search query language (SQL) is used to quickly search data and return information. Specific syntax is necessary to target data that is time consuming to find. Using SQL filter syntax such as WHERE, LIKE, OR, AND, and NOT helps searches to minimize the amount of data in the table and find what information is needed. In this project, I search through a database for specific information in a sample organization for employee and login information.

Retrieve after hours failed login attempts

My team discovered a security incident that occurred after business hours which can potentially be a brute force attempt. In this screenshot below, I use SQL filters to query the organization's database for failed login attempts. To search through all data I use **SELECT *** and FROM log_in_attempts to source the data from the specific information logs in the database. I use the **WHERE** argument to search for the after hour login time which is designated to be **18:00** or 6 o'clock. I use the **AND** to add an additional argument and use **= 0** to signal that logins were unsuccessful.

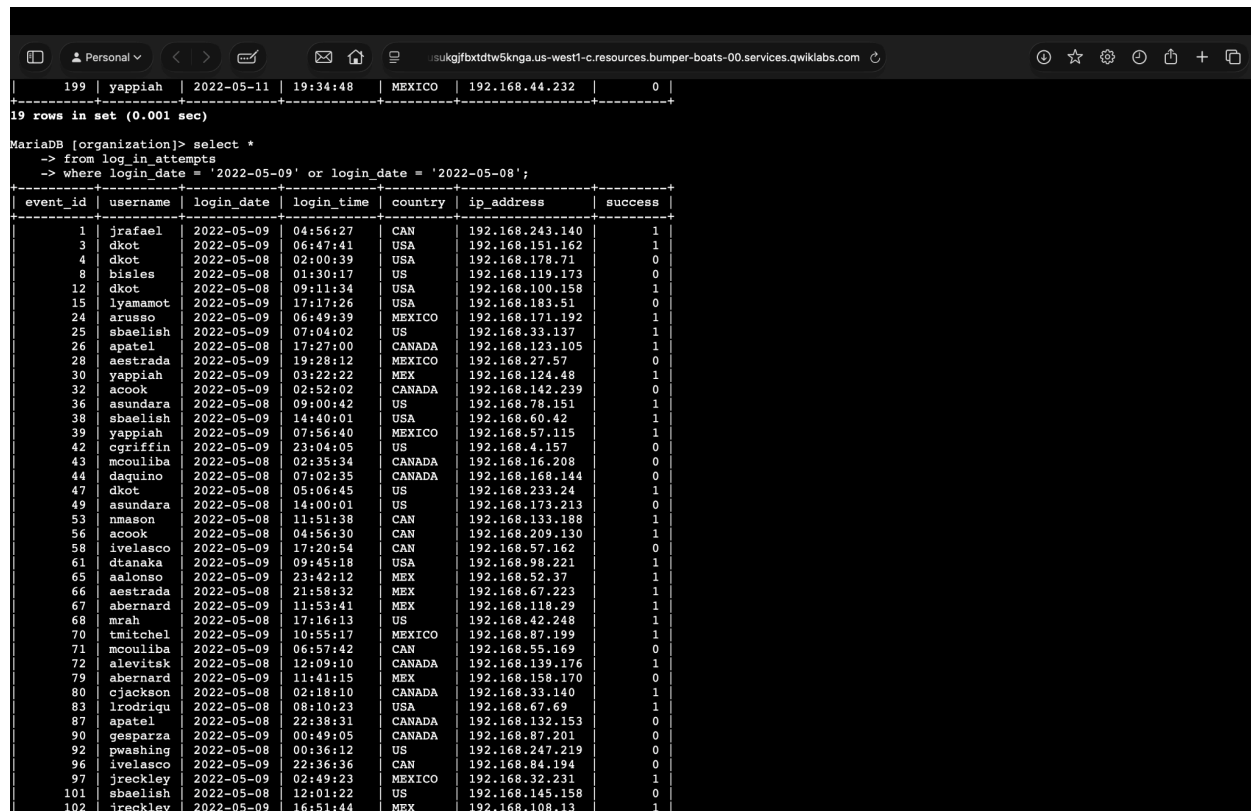
```
105 | c.jackson | 2022-05-12 | 19:36:42 | CAN | 192.168.247.153 | 1
107 | bisles | 2022-05-12 | 20:25:57 | USA | 192.168.116.187 | 0
108 | daquino | 2022-05-09 | 21:30:48 | CANADA | 192.168.15.110 | 1
111 | aestrada | 2022-05-10 | 22:00:26 | MEXICO | 192.168.76.27 | 0
115 | ivelasco | 2022-05-10 | 23:06:01 | CAN | 192.168.154.1 | 1
116 | tmitchel | 2022-05-10 | 20:33:27 | MEXICO | 192.168.119.26 | 1
118 | smartell | 2022-05-12 | 23:21:31 | MEXICO | 192.168.173.196 | 1
119 | tmitchel | 2022-05-11 | 23:07:13 | MEXICO | 192.168.110.175 | 1
121 | btang | 2022-05-10 | 22:00:36 | US | 192.168.80.143 | 1
126 | jrafael | 2022-05-12 | 18:47:52 | CAN | 192.168.22.16 | 1
127 | abellmas | 2022-05-09 | 21:20:51 | CANADA | 192.168.70.122 | 0
131 | bisles | 2022-05-09 | 20:03:55 | US | 192.168.113.171 | 0
132 | rjensen | 2022-05-12 | 23:26:03 | MEX | 192.168.9.166 | 1
155 | cgriffin | 2022-05-12 | 22:18:42 | USA | 192.168.236.176 | 0
158 | smartell | 2022-05-09 | 19:30:32 | MEXICO | 192.168.190.178 | 1
160 | jclark | 2022-05-10 | 20:49:00 | CANADA | 192.168.214.49 | 0
164 | jclark | 2022-05-12 | 21:15:52 | CAN | 192.168.18.34 | 1
173 | asundara | 2022-05-12 | 23:17:52 | US | 192.168.58.217 | 1
199 | yappiah | 2022-05-11 | 19:34:48 | MEXICO | 192.168.44.232 | 0
39 rows in set (0.001 sec)

MariaDB [organization]> select * from log_in_attempts where login_time > '18:00' and success = 0;
+-----+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address | success |
+-----+-----+-----+-----+-----+-----+-----+
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.12 | 0
| 18 | pwashing | 2022-05-11 | 19:28:50 | US | 192.168.66.142 | 0
| 20 | tshah | 2022-05-12 | 18:56:36 | MEXICO | 192.168.109.50 | 0
| 28 | aestrada | 2022-05-09 | 19:28:12 | MEXICO | 192.168.27.57 | 0
| 34 | drosas | 2022-05-11 | 21:02:04 | US | 192.168.45.93 | 0
| 42 | cgriffin | 2022-05-09 | 23:04:05 | US | 192.168.4.157 | 0
| 52 | c.jackson | 2022-05-10 | 22:07:07 | CAN | 192.168.58.27 | 0
| 69 | wjaffrey | 2022-05-11 | 19:55:15 | USA | 192.168.100.17 | 0
| 82 | abernard | 2022-05-12 | 23:38:46 | MEX | 192.168.234.49 | 0
| 87 | apatel | 2022-05-08 | 22:38:31 | CANADA | 192.168.132.153 | 0
| 96 | ivelasco | 2022-05-09 | 22:36:36 | CAN | 192.168.84.194 | 0
| 104 | asundara | 2022-05-11 | 18:38:07 | US | 192.168.96.200 | 0
| 107 | bisles | 2022-05-12 | 20:25:57 | USA | 192.168.116.187 | 0
| 111 | aestrada | 2022-05-10 | 22:00:26 | MEXICO | 192.168.76.27 | 0
| 127 | abellmas | 2022-05-09 | 21:20:51 | CANADA | 192.168.70.122 | 0
| 131 | bisles | 2022-05-09 | 20:03:55 | US | 192.168.113.171 | 0
| 155 | cgriffin | 2022-05-12 | 22:18:42 | USA | 192.168.236.176 | 0
| 160 | jclark | 2022-05-10 | 20:49:00 | CANADA | 192.168.214.49 | 0
| 199 | yappiah | 2022-05-11 | 19:34:48 | MEXICO | 192.168.44.232 | 0
19 rows in set (0.001 sec)

MariaDB [organization]>
```

Retrieve login attempts on specific dates

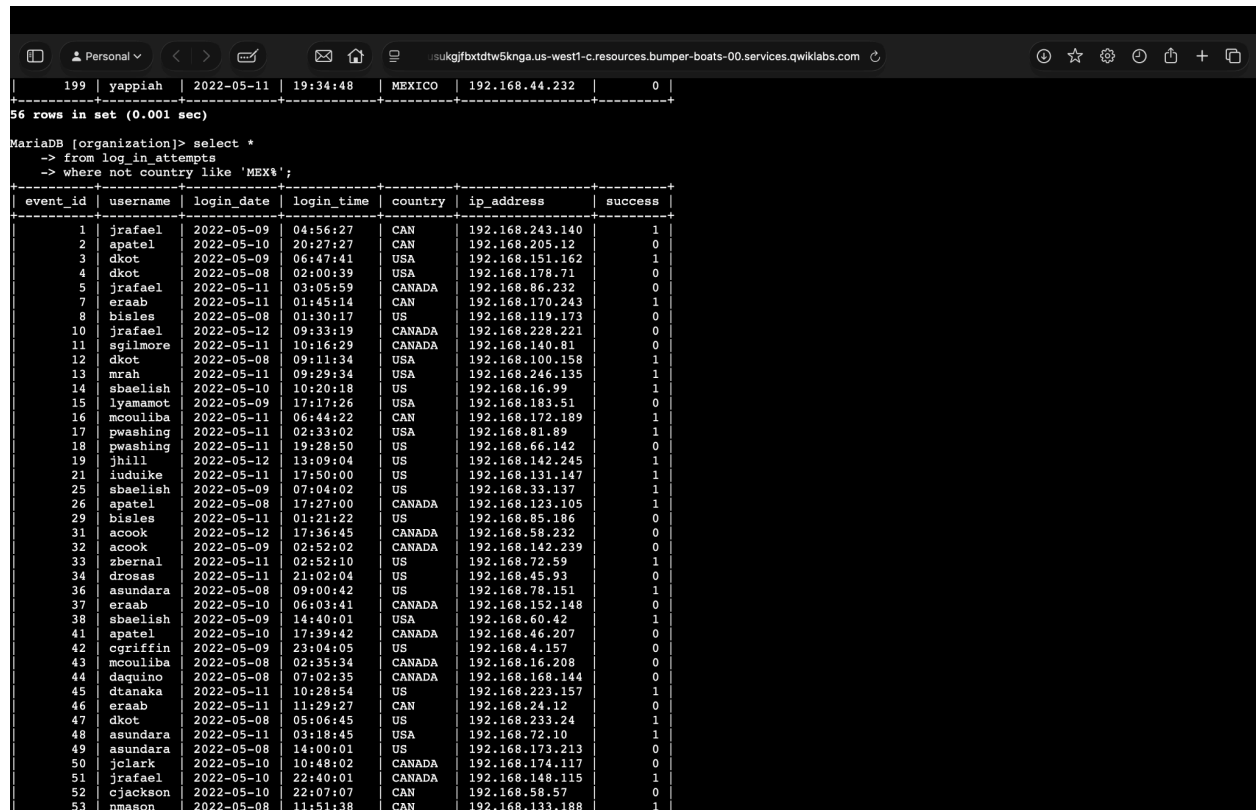
A suspicious event occurred on the date 2022-05-09. To investigate the event, login attempts for the day of the event and the day before should be reviewed. In this screenshot below, I queried for login attempts on specific dates. Again I used **SELECT *** to access data categories and **FROM log_in_attempts** for login information. Then, I searched **WHERE login_date = '2022-05-09'** and the **OR** argument for **login_date '2022-05-08'**. This gave all data for the both dates inside of the SQL search while showing whether they were successful or not.



199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0
19 rows in set (0.001 sec)						
MariaDB [organization]> select * -> from log_in_attempts -> where login_date = '2022-05-09' or login_date = '2022-05-08';						
event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
39	yappiah	2022-05-09	07:56:40	MEXICO	192.168.57.115	1
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
49	asundara	2022-05-08	14:00:01	US	192.168.173.213	0
53	nmason	2022-05-08	11:51:38	CAN	192.168.133.188	1
56	acook	2022-05-08	04:56:30	CAN	192.168.209.130	1
58	ivelasco	2022-05-09	17:20:54	CAN	192.168.57.162	0
61	dtanaka	2022-05-09	09:45:18	USA	192.168.98.221	1
65	aalonso	2022-05-09	23:42:12	MEX	192.168.52.37	1
66	aestrada	2022-05-08	21:58:32	MEX	192.168.67.223	1
67	abernard	2022-05-09	11:53:41	MEX	192.168.118.29	1
68	mrab	2022-05-08	17:16:13	US	192.168.42.248	1
70	tmitchel	2022-05-09	10:55:17	MEXICO	192.168.87.199	1
71	mcouliba	2022-05-09	06:57:42	CAN	192.168.55.169	0
72	alevitsk	2022-05-08	12:09:10	CANADA	192.168.139.176	1
79	abernard	2022-05-09	11:41:15	MEX	192.168.158.170	0
80	cjackson	2022-05-08	02:18:10	CANADA	192.168.33.140	1
83	lrodriqu	2022-05-08	08:10:23	USA	192.168.67.69	1
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
90	gesparza	2022-05-09	00:49:05	CANADA	192.168.87.201	0
92	psashing	2022-05-08	00:36:12	US	192.168.247.219	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
97	jreckley	2022-05-09	02:49:23	MEXICO	192.168.32.231	1
101	sbaelish	2022-05-08	12:01:22	US	192.168.145.158	0
102	jreckley	2022-05-09	16:51:44	MEX	192.168.108.13	1

Retrieve login attempts outside of Mexico

In this search, I searched for suspicious activity occurring outside of Mexico. To lookup this data I used **SELECT *** and **FROM log_in_attempts**. Then, I used the WHERE argument to search for countries that aren't Mexico. I added the argument **NOT country LIKE 'MEX%'**; The **NOT** argument negates the countries search which returns the opposite of the argument. The **LIKE** argument allows for searches that may contain the string starting with **'MEX'**. The **%** is positioned after the **MEX** in case that the country only uses the first 3 letters or the entire name depending on the spelling.



```
199 | yappiah | 2022-05-11 | 19:34:48 | MEXICO | 192.168.44.232 | 0 |
56 rows in set (0.001 sec)

MariaDB [organization]> select *
-> from log_in_attempts
-> where not country like 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
6	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
7	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
13	mrah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
21	iudulike	2022-05-11	17:50:00	US	192.168.131.147	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
41	apatel	2022-05-10	17:39:42	CANADA	192.168.46.207	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
45	dtanaka	2022-05-11	10:28:54	US	192.168.223.157	1
46	eraab	2022-05-11	11:29:27	CAN	192.168.24.12	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
48	asundara	2022-05-11	03:18:45	USA	192.168.72.10	1
49	asundara	2022-05-08	14:00:01	US	192.168.173.213	0
50	jelark	2022-05-10	10:48:02	CANADA	192.168.174.117	0
51	jrafael	2022-05-10	22:40:01	CANADA	192.168.148.115	1
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
53	nmason	2022-05-08	11:51:38	CAN	192.168.133.188	1

Retrieve employees in Marketing

My team wanted to perform security updates on specific employee machines in the Marketing department. I was responsible for getting information on these employee machines and will need to query the **employees** table. In this search, I used filters in SQL to create a query that identifies all employees in the Marketing department for all offices in the East building. This SQL filter required **SELECT *** and **FROM employees** to gather employee information. Then, I added an argument **WHERE department = 'Marketing'**. This argument searches for employees in the **Marketing** department. I added **AND** to join the two arguments together. Finally, the argument **office like 'EAST%'** searches for all offices in the east building. The % after **EAST** ensures that the search begins with east no matter the room number.

```
1063 | 1686m140n569 | lpope | Sales | East-226 |
1069 | NULL | jpark | Finance | East-110 |
1072 | u905v920w694 | esmith | Sales | East-421 |
1074 | w622x645y348 | dcoleman | Information Technology | East-126 |
1075 | x573y883z772 | fbautist | Marketing | East-267 |
1077 | z684u154z359 | ldavis | Human Resources | East-241 |
1082 | z301f659g551 | mjohnson | Information Technology | East-151 |
1085 | h339i498j269 | cperez | Sales | East-325 |
1088 | k865l965m233 | rgosh | Marketing | East-157 |
1095 | r194s893t593 | glopez | Information Technology | East-457 |
1096 | s375t538u194 | kjeffers | Information Technology | East-419 |
1101 | x701y250z303 | ichowdhu | Human Resources | East-233 |
1103 | NULL | randerss | Marketing | East-460 |
1109 | f229g533h679 | nlocklea | Sales | East-196 |
1113 | j781k420l510 | pjaimes | Human Resources | East-366 |
1121 | z628s557t397 | mrojas | Sales | East-288 |
1135 | z934g229h883 | khamamot | Information Technology | East-186 |
1140 | k982l199m839 | apatel2 | Human Resources | East-385 |
1142 | m674n127o823 | lsilva | Finance | East-440 |
1143 | n704o364p471 | sstark | Information Technology | East-282 |
1155 | z942a966b589 | zjones | Human Resources | East-122 |
1156 | a184b775c707 | dellery | Marketing | East-417 |
1159 | d881e710f732 | jshen | Finance | East-193 |
1161 | f951g408h866 | jjenkins | Information Technology | East-433 |
1163 | h679i515j339 | cwilliam | Marketing | East-216 |
1177 | w691u183x228 | aezra | Human Resources | East-190 |
1183 | b566c710d544 | lguraish | Human Resources | East-400 |
1187 | f963g637h851 | bhode | Finance | East-351 |
1192 | k570l183m949 | rlaghari | Information Technology | East-138 |
1195 | n516o853p957 | orainier | Finance | East-346 |
1199 | r520s571t459 | areyes | Human Resources | East-100 |

41 rows in set (0.001 sec)

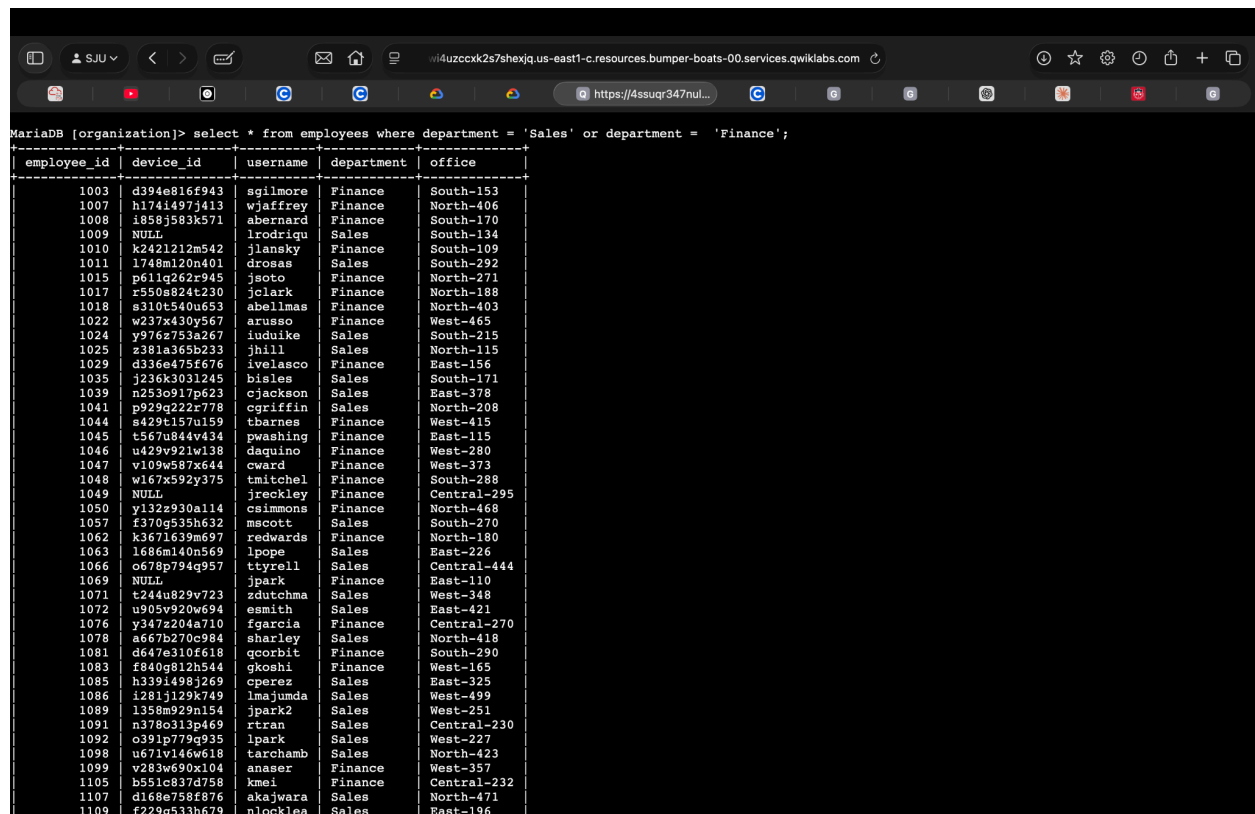
MariaDB [organization]> select *
-> from employees
-> where department = 'Marketing' and office like 'EAST%';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1000 | a320b137c219 | elarson | Marketing | East-170 |
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |
| 1075 | x573y883z772 | fbautist | Marketing | East-267 |
| 1088 | k865l965m233 | rgosh | Marketing | East-157 |
| 1103 | NULL | randerss | Marketing | East-460 |
| 1156 | a184b775c707 | dellery | Marketing | East-417 |
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |
+-----+-----+-----+-----+-----+

7 rows in set (0.033 sec)

MariaDB [organization]>
```

Retrieve employees in Finance or Sales

The security team required updates on the employee departments in the Sales and Finance departments. I inputted **SELECT *** to look through all information categories and **FROM employees** to retrieve employee information. Then, I added the argument **WHERE department = 'Sales'** to query all departments in Sales. The **OR** argument includes the 2nd part of the argument that queries the Finance department. I made sure to include **department = 'Finance'** as that includes both Sales and Finance departments.



```
MariaDB [organization]> select * from employees where department = 'Sales' or department = 'Finance';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgillmore	Finance	South-153
1007	h1741497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriagu	Sales	South-134
1010	k2421212m542	jlansky	Finance	South-109
1011	1748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1029	d336e475f676	ivelasgo	Finance	East-156
1035	j236k303l245	hisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929q222r778	cgriffin	Sales	North-208
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1057	f370g535h632	mccott	Sales	South-270
1062	k367l639m697	redwards	Finance	North-180
1063	l686m140n569	lpope	Sales	East-226
1066	o678p794q957	ttyrell	Sales	Central-444
1069	NULL	jpark	Finance	East-110
1071	t244u829v723	zdutchma	Sales	West-348
1072	u905v920w694	esmith	Sales	East-421
1076	y347z204a710	fgarcia	Finance	Central-270
1078	a667b270c984	sharley	Sales	North-418
1081	d647e310f618	qcorbit	Finance	South-290
1083	f840g812h544	gvoshi	Finance	West-165
1085	h339i498j269	cperez	Sales	East-325
1086	i281j129k749	lmajumda	Sales	West-499
1089	l358m929n154	jpark2	Sales	West-251
1091	n378o313p469	rtran	Sales	Central-230
1092	o391p779q935	lpark	Sales	West-227
1098	u671v146w618	tarchamb	Sales	North-423
1099	v283w690x104	anaser	Finance	West-357
1105	b551c837d758	kmei	Finance	Central-232
1107	d168e758f876	akajwara	Sales	North-471
1109	f229g533h679	nlocklea	Sales	East-196

Retrieve all employees not in IT

The security team required one more update for all departments excluding IT. IT had already received the update and all other departments required it. I created an SQL query beginning with **SELECT *** and **FROM employees** again. Then, I added 2 arguments **WHERE NOT** to lookup departments that are not Information Technology. The **NOT** negates the accepted argument which will return every other department.

```
33 rows in set, 5 warnings (0.001 sec)
```

MariaDB [organization]> select *
-> from employees
-> where not department = 'Information Technology';

employee_id	device_id	username	department	office
1000	a320b137c219	el Larson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodrigu	Sales	South-134
1010	k242l121m542	jlansky	Finance	South-109
1011	l748m120n401	drozas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1016	q793r736s288	sbaelish	Human Resources	North-229
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1020	u899v381w363	arutley	Marketing	South-351
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	ludulke	Sales	South-215
1025	z301a365b233	jhill	Sales	North-115
1026	a998b568c863	spatel	Human Resources	West-320
1027	b806c503d354	mrar	Marketing	West-246
1028	c603d749e374	aestrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156
1030	e391f189g913	mabadi	Marketing	West-375
1031	f419g188h578	dkot	Marketing	West-408
1034	i679j565k940	bsand	Human Resources	East-484
1035	j236k303l245	bisles	Sales	South-171
1036	k550l53m205	rjensen	Marketing	Central-239
1038	m873n636o225	btang	Human Resources	Central-260
1039	n253o917p623	cjackson	Sales	East-378
1040	o783p832q294	dtarly	Human Resources	East-237
1041	p929q822r778	cgriffin	Sales	North-208
1042	q175r338s833	acook	Human Resources	West-381
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1051	z451a308b518	itraora	Marketing	Central-134
1052	a192b174c940	jdarosa	Marketing	East-195
1053	b979c871d361	nemmanue	Human Resources	Central-259

Match employees to their machines

The department required machines to be matched to employees. This would work with an inner join which correctly returns columns that are needed based on columns. I queried this by using **SELECT *** and **FROM machines**. Then, I used the new syntax of **INNER JOIN** and matched it with the employees table. Then, I added **ON machines.device_id = employees.device_id** to specify table 1 to join table 2 which is the employees table. This returned all employee information and machines that have matches to create a bigger table. The **device_id** is a shared key which is the only information shared on both piles of data.

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
MariaDB [organization]> clear
MariaDB [organization]> clear
MariaDB [organization]> select *
-> from machines
-> inner join employees on machines.device_id = employees.device_id;
```

device_id	operating_system	email_client	OS_patch_date	employee_id	employee_id	device_id	username	department	office
a320b137c219	OS 2	Email Client 2	2021-03-01	1000	1000	a320b137c219	elarsen	Marketing	East-170
b239c825d303	OS 1	Email Client 1	2021-03-01	1001	1001	b239c825d303	bmoreno	Marketing	Central-276
c116d593e558	OS 3	Email Client 1	2021-09-01	1002	1002	c116d593e558	tshah	Human Resources	North-434
d394e816f943	OS 3	Email Client 2	2021-03-01	1003	1003	d394e816f943	sgilmore	Finance	South-153
e218f877q788	OS 2	Email Client 1	2021-09-01	1004	1004	e218f877q788	eraab	Human Resources	South-127
f551g340h864	OS 3	Email Client 2	2021-12-01	1005	1005	f551g340h864	gesparza	Human Resources	South-366
g329h357i597	OS 1	Email Client 2	2021-06-01	1006	1006	g329h357i597	alevitsk	Information Technology	East-320
h174i497j413	OS 2	Email Client 1	2021-03-01	1007	1007	h174i497j413	wjaffrey	Finance	North-406
i858j583k571	OS 2	Email Client 2	2021-06-01	1008	1008	i858j583k571	abernard	Finance	South-170
k242l212m542	OS 1	Email Client 1	2021-03-01	1010	1010	k242l212m542	jlansky	Finance	South-109
l748m120n401	OS 3	Email Client 1	2021-09-01	1011	1011	l748m120n401	droas	Sales	South-292
m756n668o146	OS 1	Email Client 2	2021-09-01	1012	1012	m756n668o146	nmason	Information Technology	North-160
n205o559p243	OS 1	Email Client 2	2021-06-01	1013	1013	n205o559p243	zbernal	Information Technology	South-229
p611q262r945	OS 1	Email Client 2	2021-03-01	1015	1015	p611q262r945	jsoto	Finance	North-271
q793r736s288	OS 2	Email Client 1	2021-03-01	1016	1016	q793r736s288	sbaelish	Human Resources	North-229
r550s824t230	OS 2	Email Client 1	2021-03-01	1017	1017	r550s824t230	jclark	Finance	North-188
s310t540u653	OS 1	Email Client 2	2021-12-01	1018	1018	s310t540u653	abellmas	Finance	North-403
t815u205v470	OS 3	Email Client 1	2021-06-01	1019	1019	t815u205v470	mcouliba	Information Technology	North-108
u899v381w363	OS 3	Email Client 1	2021-06-01	1020	1020	u899v381w363	arutley	Marketing	South-351
v200w121x977	OS 2	Email Client 2	2021-03-01	1021	1021	v200w121x977	smartell	Information Technology	South-138
w237x430y567	OS 2	Email Client 2	2021-06-01	1022	1022	w237x430y567	aruaeo	Finance	West-465
x253y759z103	OS 1	Email Client 2	2021-12-01	1023	1023	x253y759z103	aalonso	Information Technology	West-393
y976z753a267	OS 2	Email Client 2	2021-06-01	1024	1024	y976z753a267	iuduke	Sales	South-215
z381a365b233	OS 3	Email Client 2	2021-12-01	1025	1025	z381a365b233	jhill	Sales	North-115
a998b568c863	OS 3	Email Client 1	2021-12-01	1026	1026	a998b568c863	apatel	Human Resources	West-320
b806c503d354	OS 2	Email Client 1	2021-12-01	1027	1027	b806c503d354	mrah	Marketing	West-246
c603d749e374	OS 1	Email Client 1	2021-12-01	1028	1028	c603d749e374	astrada	Human Resources	West-121
d336e475f676	OS 2	Email Client 2	2021-06-01	1029	1029	d336e475f676	ivelasco	Finance	East-156
e391f189g913	OS 3	Email Client 2	2021-12-01	1030	1030	e391f189g913	mabadi	Marketing	West-375
f419g188h578	OS 1	Email Client 1	2021-06-01	1031	1031	f419g188h578	dkot	Marketing	West-408
g773h303i639	OS 2	Email Client 2	2021-12-01	1032	1032	g773h303i639	jrafasl	Information Technology	Central-309
i679j565k940	OS 1	Email Client 1	2021-03-01	1034	1034	i679j565k940	bsand	Human Resources	East-484
j236k303l245	OS 1	Email Client 1	2021-12-01	1035	1035	j236k303l245	bisles	Sales	South-171
k550l533m205	OS 3	Email Client 2	2021-09-01	1036	1036	k550l533m205	rjensen	Marketing	Central-239
l693m585n528	OS 3	Email Client 1	2021-06-01	1037	1037	l693m585n528	dtanaka	Information Technology	West-468
m873n636o225	OS 1	Email Client 2	2021-06-01	1038	1038	m873n636o225	btang	Human Resources	Central-260
n253o917p623	OS 1	Email Client 2	2021-09-01	1039	1039	n253o917p623	cjackson	Sales	East-378
o783p832q294	OS 2	Email Client 2	2021-09-01	1040	1040	o783p832q294	dtarly	Human Resources	East-237
p929q222r778	OS 3	Email Client 1	2021-06-01	1041	1041	p929q222r778	cgriffin	Sales	North-208
q175r338s833	OS 3	Email Client 1	2021-06-01	1042	1042	q175r338s833	acook	Human Resources	West-381
s429t157u159	OS 1	Email Client 1	2021-03-01	1044	1044	s429t157u159	tbarnes	Finance	West-415

Return information on employees and machines

The organization asks me to find every device that is unassigned in the closet. If a machine has no user then it will return as null. A left join is where every row from the first table is kept even if there is no second table. Again, I begin the query with SELECT * and FROM machines. I use a LEFT JOIN on employees using device_id from both machines and employees table. This returns every single employee regardless if they were assigned a device.

```
MariaDB [organization]> select *
-> from machines
-> left join employees on machines.device_id = employees.device_id;
```

device_id	operating_system	email_client	OS_patch_date	employee_id	employee_id	device_id	username	department	office
a320b137c219	OS 2	Email Client 2	2021-03-01	1000	1000	a320b137c219	elarson	Marketing	East-170
b239c825d303	OS 1	Email Client 1	2021-03-01	1001	1001	b239c825d303	bmoreno	Marketing	Central-276
c116d593e558	OS 3	Email Client 1	2021-09-01	1002	1002	c116d593e558	tshah	Human Resources	North-434
d394e916f943	OS 3	Email Client 2	2021-03-01	1003	1003	d394e916f943	sgilmore	Finance	South-153
e218f877g788	OS 2	Email Client 1	2021-09-01	1004	1004	e218f877g788	eraab	Human Resources	South-127
f551g340h864	OS 3	Email Client 2	2021-12-01	1005	1005	f551g340h864	gesparza	Human Resources	South-366
g329h357i597	OS 1	Email Client 2	2021-06-01	1006	1006	g329h357i597	alevitsk	Information Technology	East-320
h174i497j413	OS 2	Email Client 1	2021-03-01	1007	1007	h174i497j413	wjaffrey	Finance	North-406
i858j583k571	OS 2	Email Client 2	2021-06-01	1008	1008	i858j583k571	abernard	Finance	South-170
k242l121m542	OS 1	Email Client 1	2021-03-01	1010	1010	k242l121m542	jlansky	Finance	South-109
l748m120n401	OS 3	Email Client 1	2021-09-01	1011	1011	l748m120n401	drosas	Sales	South-292
m756n668o146	OS 1	Email Client 2	2021-09-01	1012	1012	m756n668o146	nmason	Information Technology	North-160
n205o559p243	OS 1	Email Client 2	2021-06-01	1013	1013	n205o559p243	zbernal	Information Technology	South-229
p611q262r945	OS 1	Email Client 2	2021-03-01	1015	1015	p611q262r945	jsoto	Finance	North-271
q793r736s288	OS 2	Email Client 1	2021-03-01	1016	1016	q793r736s288	sbaelish	Human Resources	North-229
r550s824t230	OS 2	Email Client 1	2021-03-01	1017	1017	r550s824t230	jclark	Finance	North-188
s310t540u653	OS 1	Email Client 2	2021-12-01	1018	1018	s310t540u653	abellmas	Finance	North-403
t815u205v470	OS 3	Email Client 1	2021-06-01	1019	1019	t815u205v470	mcouliba	Information Technology	North-108
u899v381w363	OS 3	Email Client 1	2021-06-01	1020	1020	u899v381w363	arutley	Marketing	South-351
v200w121x977	OS 2	Email Client 2	2021-03-01	1021	1021	v200w121x977	smartell	Information Technology	South-138
w237x430y567	OS 2	Email Client 2	2021-06-01	1022	1022	w237x430y567	arusso	Finance	West-465
x253y759z103	OS 1	Email Client 2	2021-12-01	1023	1023	x253y759z103	aalonso	Information Technology	West-393
y976z753a267	OS 2	Email Client 2	2021-06-01	1024	1024	y976z753a267	iudulke	Sales	South-215
z381a365b233	OS 3	Email Client 2	2021-12-01	1025	1025	z381a365b233	jhill	Sales	North-115
a998b568c863	OS 3	Email Client 1	2021-12-01	1026	1026	a998b568c863	spatel	Human Resources	West-320
b806c503d354	OS 2	Email Client 1	2021-12-01	1027	1027	b806c503d354	mrah	Marketing	West-246
c603d749e374	OS 1	Email Client 1	2021-12-01	1028	1028	c603d749e374	aestrada	Human Resources	West-121
d336e475f676	OS 2	Email Client 2	2021-06-01	1029	1029	d336e475f676	ivelasco	Finance	East-156
e391f189g913	OS 3	Email Client 2	2021-12-01	1030	1030	e391f189g913	mabadi	Marketing	West-375
f419g188h578	OS 1	Email Client 1	2021-06-01	1031	1031	f419g188h578	dkot	Marketing	West-408
g773h303i639	OS 2	Email Client 2	2021-12-01	1032	1032	g773h303i639	jrafael	Information Technology	Central-309
i679j565k940	OS 1	Email Client 1	2021-03-01	1034	1034	i679j565k940	bsand	Human Resources	East-484
j236k303l245	OS 1	Email Client 1	2021-12-01	1035	1035	j236k303l245	bisles	Sales	South-171
k550l533m205	OS 3	Email Client 2	2021-09-01	1036	1036	k550l533m205	rjensen	Marketing	Central-239
l693m585n528	OS 3	Email Client 1	2021-06-01	1037	1037	l693m585n528	dtanaka	Information Technology	West-468
m873n636o225	OS 1	Email Client 2	2021-06-01	1038	1038	m873n636o225	btang	Human Resources	Central-260
n253o917p623	OS 1	Email Client 2	2021-09-01	1039	1039	n253o917p623	cjackson	Sales	East-378
o783p832q294	OS 2	Email Client 2	2021-09-01	1040	1040	o783p832q294	dtarly	Human Resources	East-237
p929q222r778	OS 3	Email Client 1	2021-06-01	1041	1041	p929q222r778	cgriffin	Sales	North-208
q175r338s833	OS 3	Email Client 1	2021-06-01	1042	1042	q175r338s833	acook	Human Resources	West-381
s429t157u159	OS 1	Email Client 1	2021-03-01	1044	1044	s429t157u159	tbarnes	Finance	West-415
t567u844v434	OS 3	Email Client 2	2021-12-01	1045	1045	t567u844v434	pwashing	Finance	East-115
u429v921w138	OS 2	Email Client 2	2021-12-01	1046	1046	u429v921w138	daquino	Finance	West-280
v109w587x644	OS 2	Email Client 2	2021-03-01	1047	1047	v109w587x644	cward	Finance	West-373

The organization asks to see all employees on even if they haven't been issued a laptop. If an employee has no machine then a null value will return from the lookup. A right join keeps every single row on the second table even if there is no match on the first. In this query, I used a right join on employees. I began the query with **SELECT *** and **FROM machines**. Then, I added the **RIGHT JOIN** employees syntax **ON machines.device_id**. I made it **= to employees.device_id** which matched the shared row. This returned all employees regardless of if they have a device or not.

MariaDB [organization]> select *
-> from machines
-> right join employees on machines.device_id = employees.device_id;

device_id	operating_system	email_client	OS_patch_date	employee_id	employee_id	device_id	username	department	office
a320b137c219	OS 2	Email Client 2	2021-03-01	1000	1000	a320b137c219	elarson	Marketing	East-170
b239c825d303	OS 1	Email Client 1	2021-03-01	1001	1001	b239c825d303	bmoreno	Marketing	Central-276
c116d593e558	OS 3	Email Client 1	2021-09-01	1002	1002	c116d593e558	tshah	Human Resources	North-434
d394e816f943	OS 3	Email Client 2	2021-03-01	1003	1003	d394e816f943	sgilmore	Finance	South-153
e218f877g788	OS 2	Email Client 1	2021-09-01	1004	1004	e218f877g788	eraab	Human Resources	South-127
f551g340h864	OS 3	Email Client 2	2021-12-01	1005	1005	f551g340h864	gesparza	Human Resources	South-366
g329h357i597	OS 1	Email Client 2	2021-06-01	1006	1006	g329h357i597	alevitsk	Information Technology	East-320
h174i497j413	OS 2	Email Client 1	2021-03-01	1007	1007	h174i497j413	wjaffrey	Finance	North-406
i858j583k571	OS 2	Email Client 2	2021-06-01	1008	1008	i858j583k571	abernard	Finance	South-170
NULL	NULL	NULL	NULL	1009	1009	NULL	lrodriagu	Sales	South-134
k242l212m542	OS 1	Email Client 1	2021-03-01	1010	1010	k242l212m542	jlansky	Finance	South-109
l748m120n401	OS 3	Email Client 1	2021-09-01	1011	1011	l748m120n401	drozas	Sales	South-292
m756n668o146	OS 1	Email Client 2	2021-09-01	1012	1012	m756n668o146	nmason	Information Technology	North-160
n205o559p243	OS 1	Email Client 2	2021-06-01	1013	1013	n205o559p243	zbernal	Information Technology	South-229
NULL	NULL	NULL	NULL	1014	1014	NULL	asundara	Information Technology	West-219
p611q262r945	OS 1	Email Client 2	2021-03-01	1015	1015	p611q262r945	jsoto	Finance	North-271
q793r736s288	OS 2	Email Client 1	2021-03-01	1016	1016	q793r736s288	sbaelish	Human Resources	North-229
r550s824t230	OS 2	Email Client 1	2021-03-01	1017	1017	r550s824t230	jclark	Finance	North-188
s310t540u653	OS 1	Email Client 2	2021-12-01	1018	1018	s310t540u653	abellmas	Finance	North-403
t815u205v470	OS 3	Email Client 1	2021-06-01	1019	1019	t815u205v470	mcouliba	Information Technology	North-108
u899v381w363	OS 3	Email Client 1	2021-06-01	1020	1020	u899v381w363	arutley	Marketing	South-351
v200w121x977	OS 2	Email Client 2	2021-03-01	1021	1021	v200w121x977	smartell	Information Technology	South-138
w237x430y567	OS 2	Email Client 2	2021-06-01	1022	1022	w237x430y567	arusso	Finance	West-465
x253y759z103	OS 1	Email Client 2	2021-12-01	1023	1023	x253y759z103	aalonso	Information Technology	West-393
y976z753a267	OS 2	Email Client 2	2021-06-01	1024	1024	y976z753a267	iuduike	Sales	South-215
z381a365b233	OS 3	Email Client 2	2021-12-01	1025	1025	z381a365b233	jhill	Sales	North-115
a998b568c863	OS 3	Email Client 1	2021-12-01	1026	1026	a998b568c863	apatel	Human Resources	West-320
b806c503d354	OS 2	Email Client 1	2021-12-01	1027	1027	b806c503d354	mrah	Marketing	West-246
c603d749e374	OS 1	Email Client 1	2021-12-01	1028	1028	c603d749e374	aestrada	Human Resources	West-121
d336e475f676	OS 2	Email Client 2	2021-06-01	1029	1029	d336e475f676	ivelasco	Finance	East-156
e391f189g913	OS 3	Email Client 2	2021-12-01	1030	1030	e391f189g913	mabadi	Marketing	West-375
f419g188h578	OS 1	Email Client 1	2021-06-01	1031	1031	f419g188h578	dkot	Marketing	West-408
g773h303i639	OS 2	Email Client 2	2021-12-01	1032	1032	g773h303i639	jrafael	Information Technology	Central-309
NULL	NULL	NULL	NULL	1033	1033	NULL	yappiah	Information Technology	West-387
i679j565k940	OS 1	Email Client 1	2021-03-01	1034	1034	i679j565k940	bsand	Human Resources	East-484
j236k303l245	OS 1	Email Client 1	2021-12-01	1035	1035	j236k303l245	bisles	Sales	South-171
k550l533m205	OS 3	Email Client 2	2021-09-01	1036	1036	k550l533m205	rjensen	Marketing	Central-239
l693m585n528	OS 3	Email Client 1	2021-06-01	1037	1037	l693m585n528	dtanaka	Information Technology	West-468
m873n636o225	OS 1	Email Client 2	2021-06-01	1038	1038	m873n636o225	btang	Human Resources	Central-260
n253o917p623	OS 1	Email Client 2	2021-09-01	1039	1039	n253o917p623	cjackson	Sales	East-378
o782p932q294	OS 2	Email Client 2	2021-09-01	1040	1040	o782p932q294	dtarly	Human Resources	East-237
p929q222r778	OS 3	Email Client 1	2021-06-01	1041	1041	p929q222r778	cgriffin	Sales	North-208
q175r338s833	OS 3	Email Client 1	2021-06-01	1042	1042	q175r338s833	acook	Human Resources	West-381
NULL	NULL	NULL	NULL	1043	1043	NULL	lyamamot	Information Technology	East-354

Retrieve login attempt data

To continue the investigation on the security incident, a query regarding information on all employees who made login attempts was required. An inner join was used for this process. I began the query with **SELECT *** and **FROM employees**. Then, I added **INNER JOIN log_in_attempts** to garner data on login attempts. Then, I added on **employees.username = log_in_attempts.username** to link common usernames on the database. This allowed me to view all IP addresses, successful logins, and the username of the user that logged in at a specific time.

Personal

Summary

Overall, this was a great introduction experience into SQL and database queries. Using different arguments to filter each query gave insight into how real-world database environments are utilized. The different arguments have purposes and garner information uniquely. This showed the significance of data efficiency as without adding filters would make data lookup time consuming.